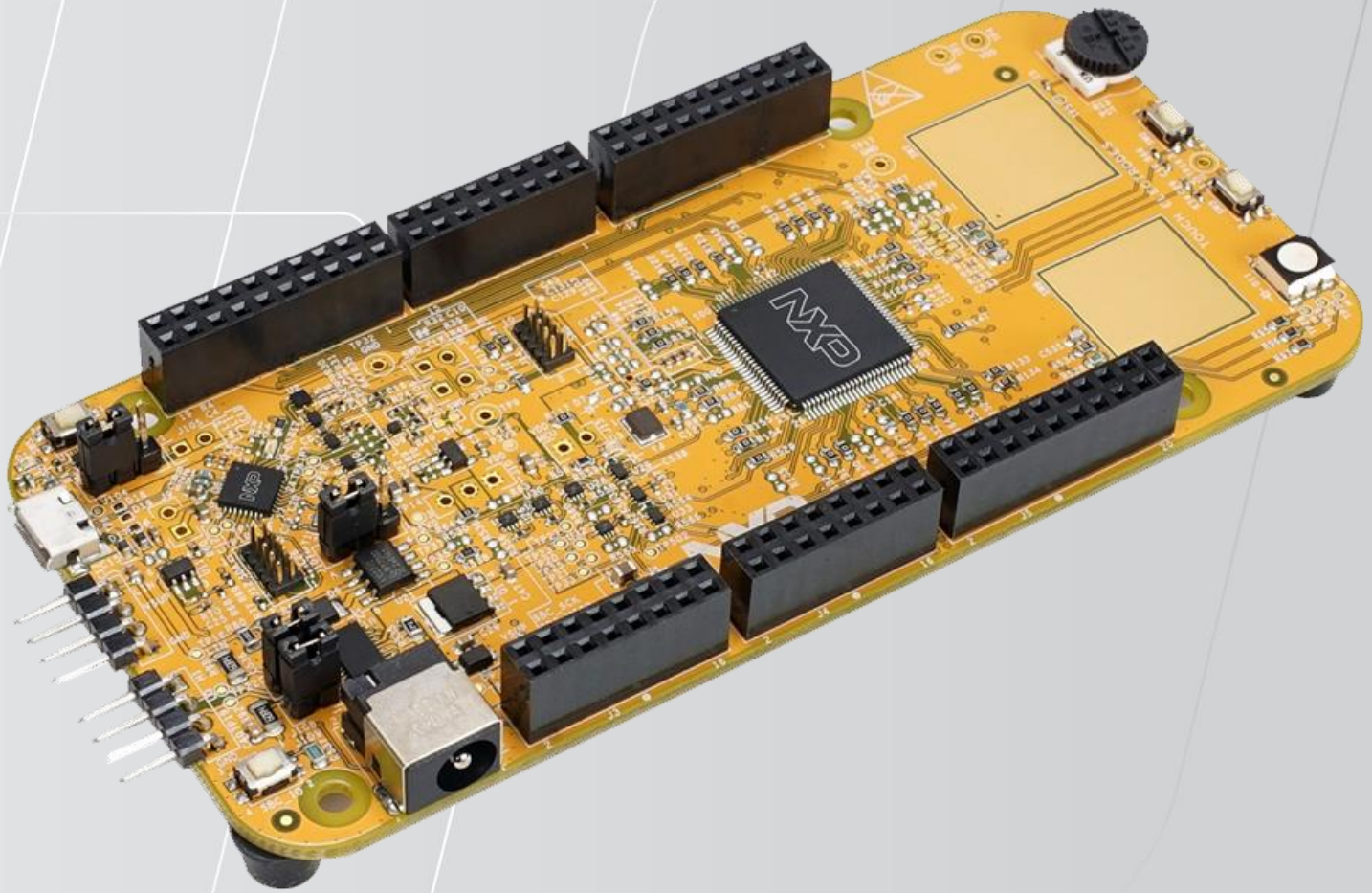
SPI



10 Dec 2022

OUTLINE

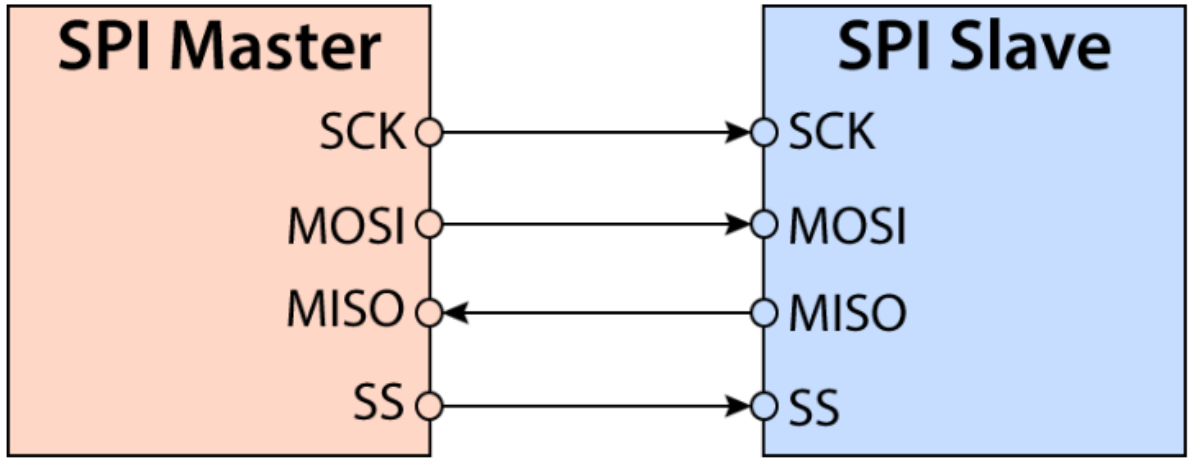
1. SPI overview
2. LPSPI
3. Assignment

1. SPI overview

- The Serial Peripheral Interface (SPI) bus is a synchronous serial communication interface used for short distance communication between various peripheral devices. The SPI bus interconnects a single master device with one or more slave devices.
- The master device originates the data transmission by selecting a slave device and generating a clock. Data is transmitted in both directions simultaneously (full-duplex mode) or in one direction (half-duplex mode)
- SPI is developed by Motorola
- Communicate with MCU, sensors, EEPROM, LCD, SD card,...

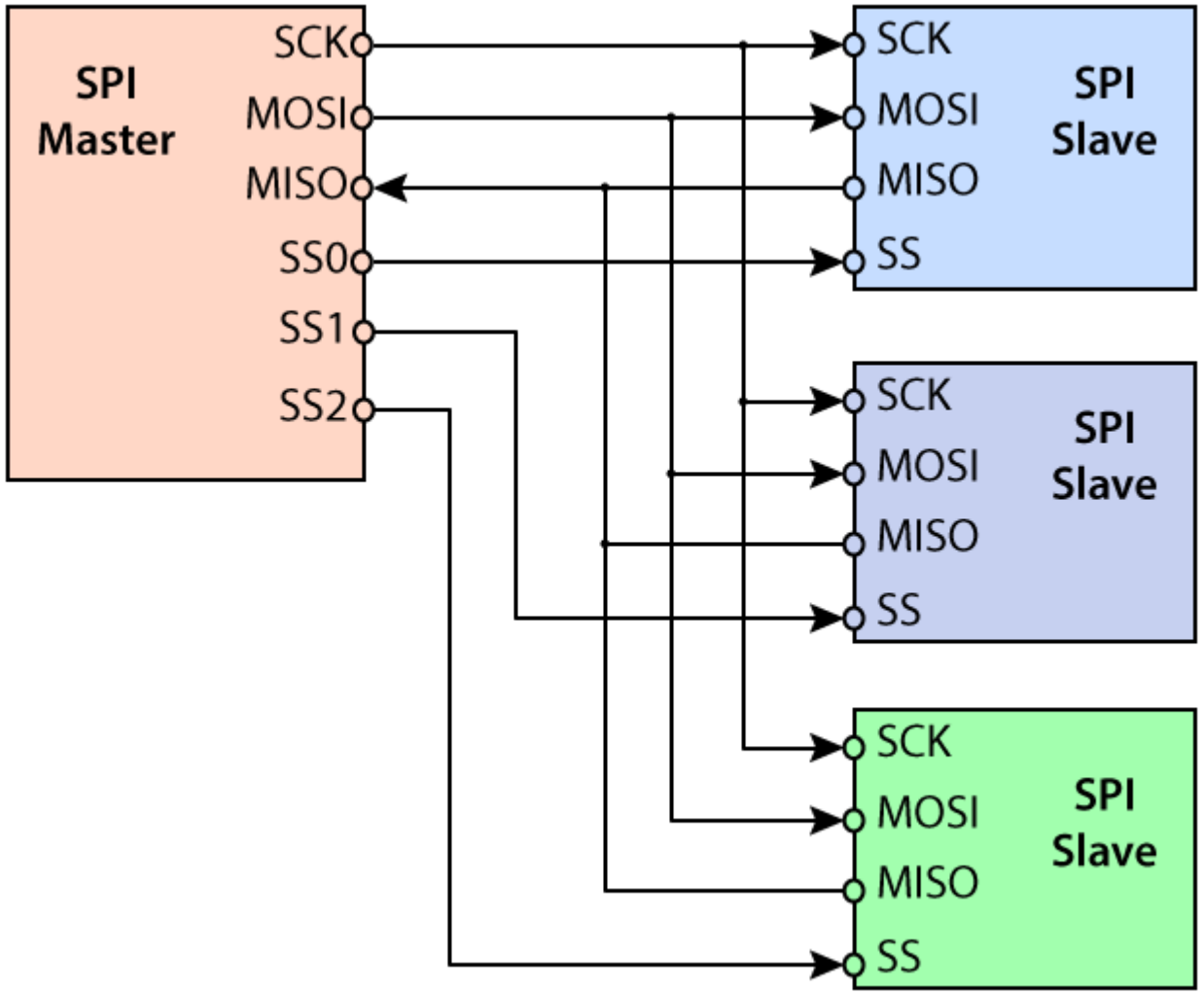
1.1. SPI Bus Structure

The SPI bus uses four signal lines:
- SCK (Clock) – The master generates the clock to synchronize data transmission
- MOSI (Master Output, Slave Input) – The master sends data on the MOSI line, the slave receives it
- MISO (Master Input, Slave Output) – The slave sends data on the MISO line, the master receives it
- CS or SS (Chip Select or Slave Select) – The master drops the SS line to select the specific SPI slave device.



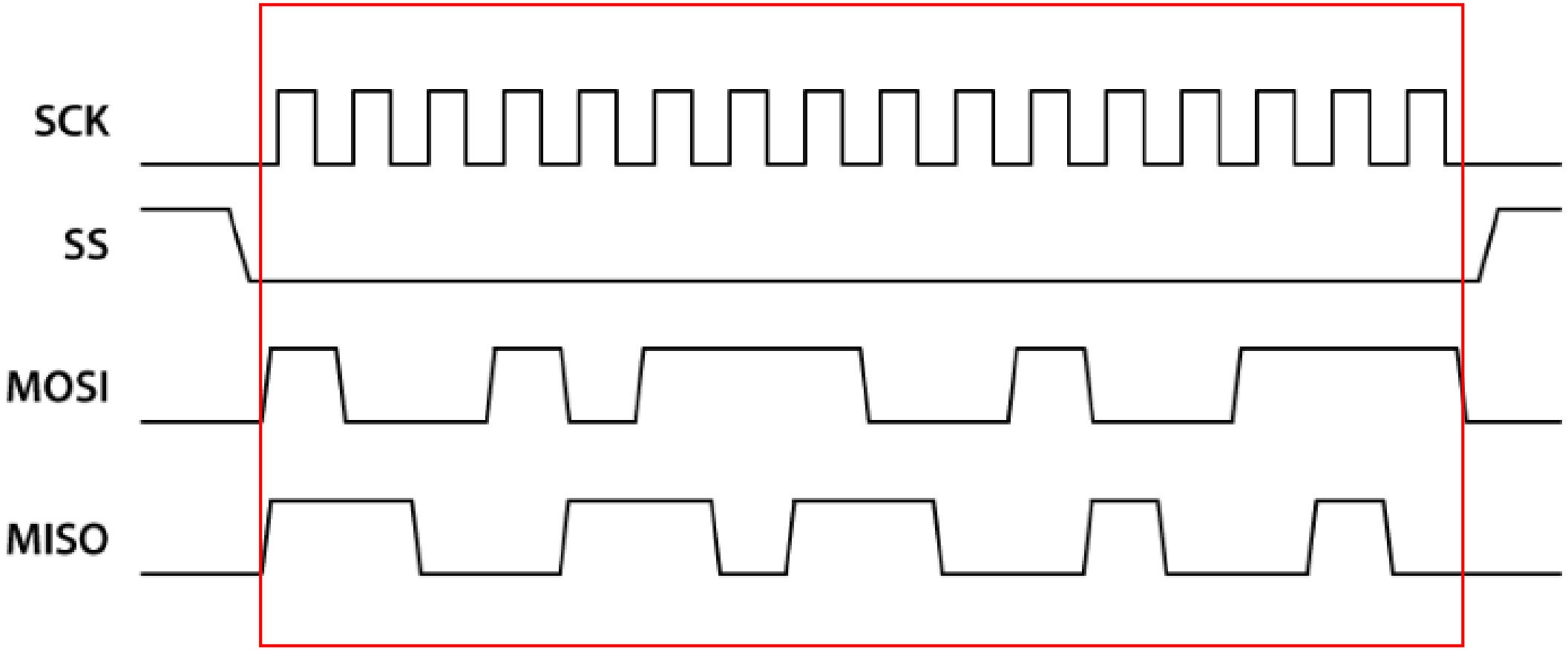
The SPI bus with a single master and a single slave

1.1. SPI Bus Structure

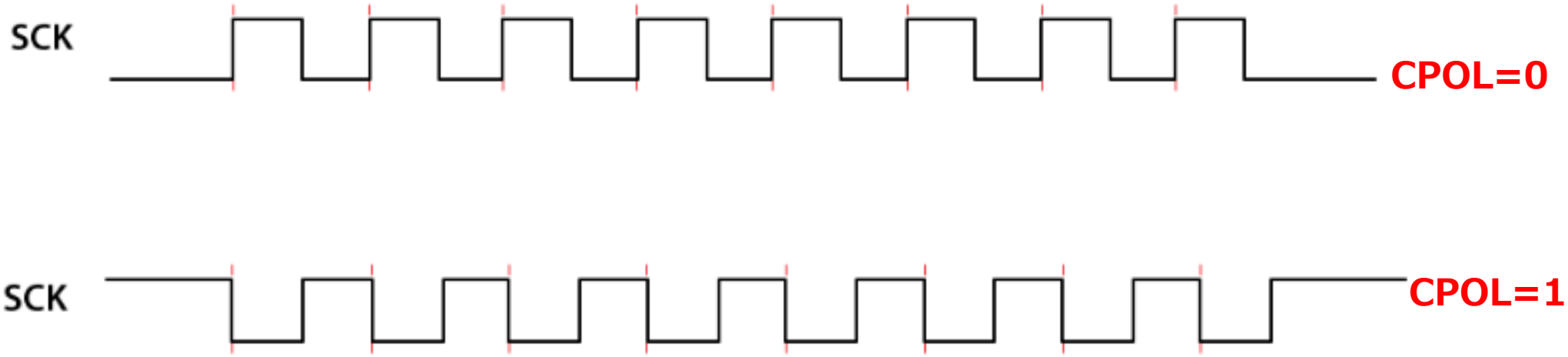


The SPI bus with a single master and multiple slaves

1.2. SPI Data Frames



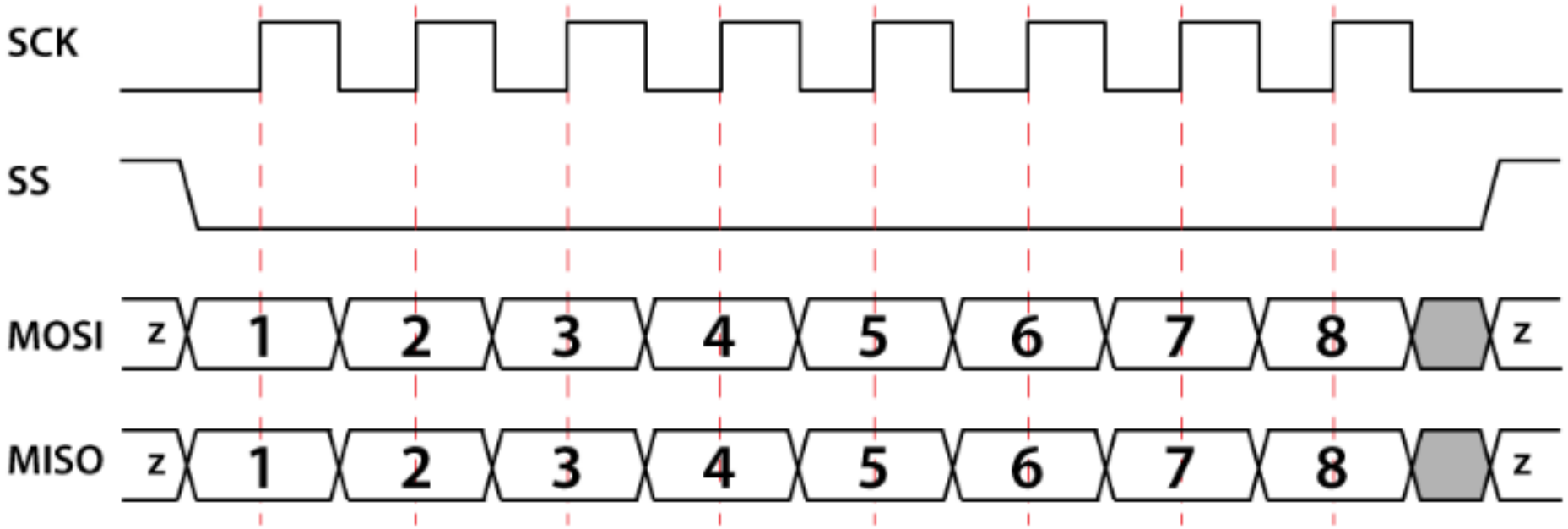
1.3. Clock Phase and Polarity



The **CPOL** parameter assigns the clock level when the clock is not active

The clock (SCK) signal may be inverted (CPOL=1) or non-inverted (CPOL=0). For the inverted clock signal, the first clock edge is falling. For the non-inverted clock signal, the first clock edge is rising.

1.3. Clock Phase and Polarity



The **CPHA** parameter is used to shift the capturing phase.

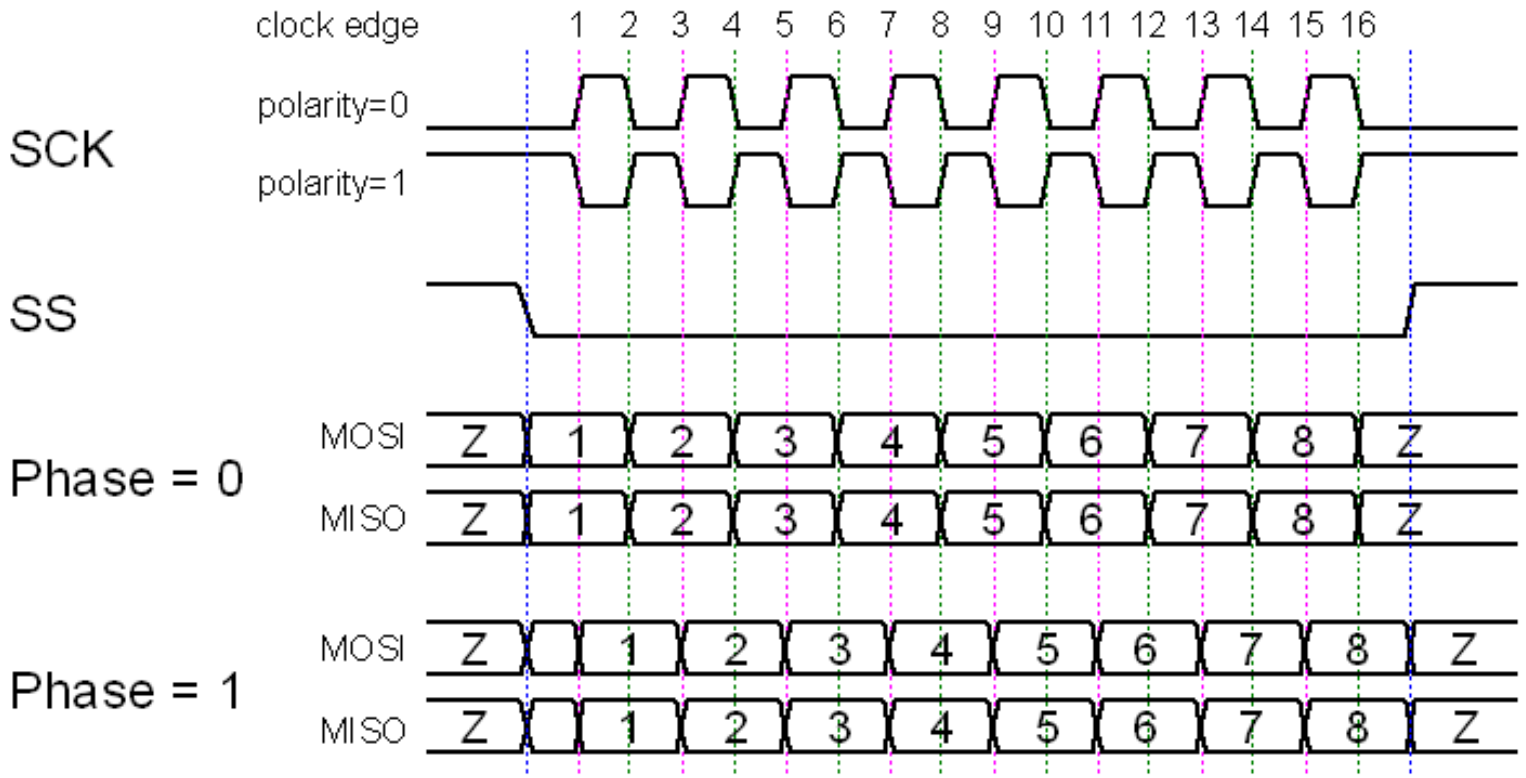
If CPHA=0, the data are captured on the leading (first) clock edge.

If CPHA=1, the data are captured on the trailing (second) clock edge.

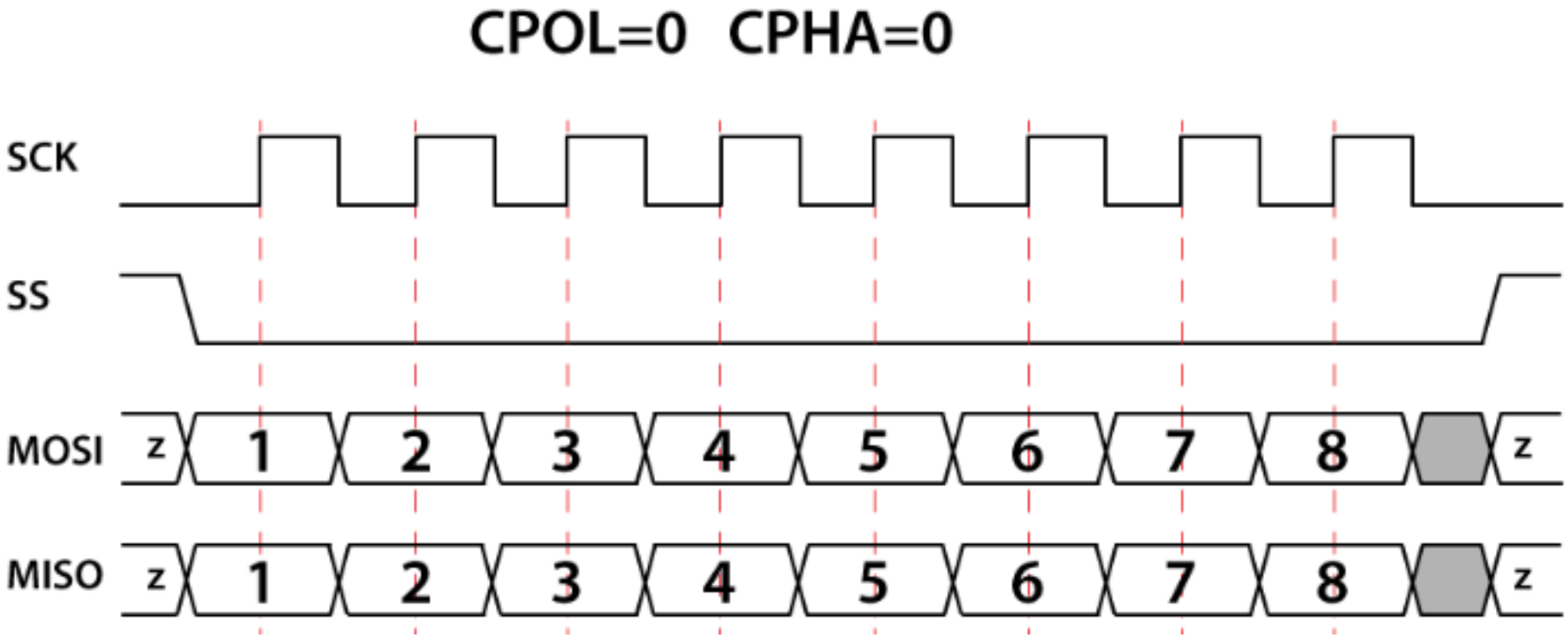
1.3. Clock Phase and Polarity

There are four possible modes that can be used in an SPI protocol:

1. CPOL = 0, CPHA = 0
2. CPOL = 0, CPHA = 1
3. CPOL = 1, CPHA = 0
4. CPOL = 1, CPHA = 1

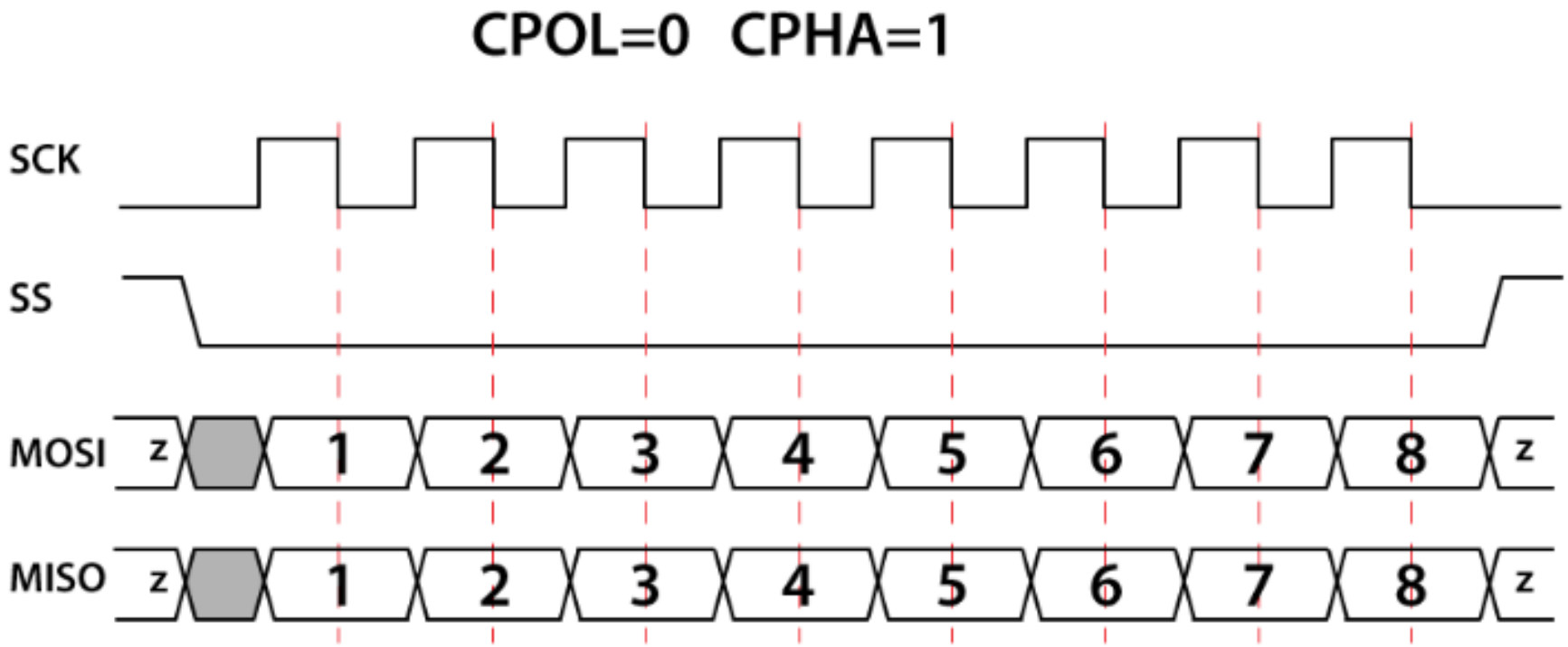


1.3.1. CPOL = 0, CPHA = 0



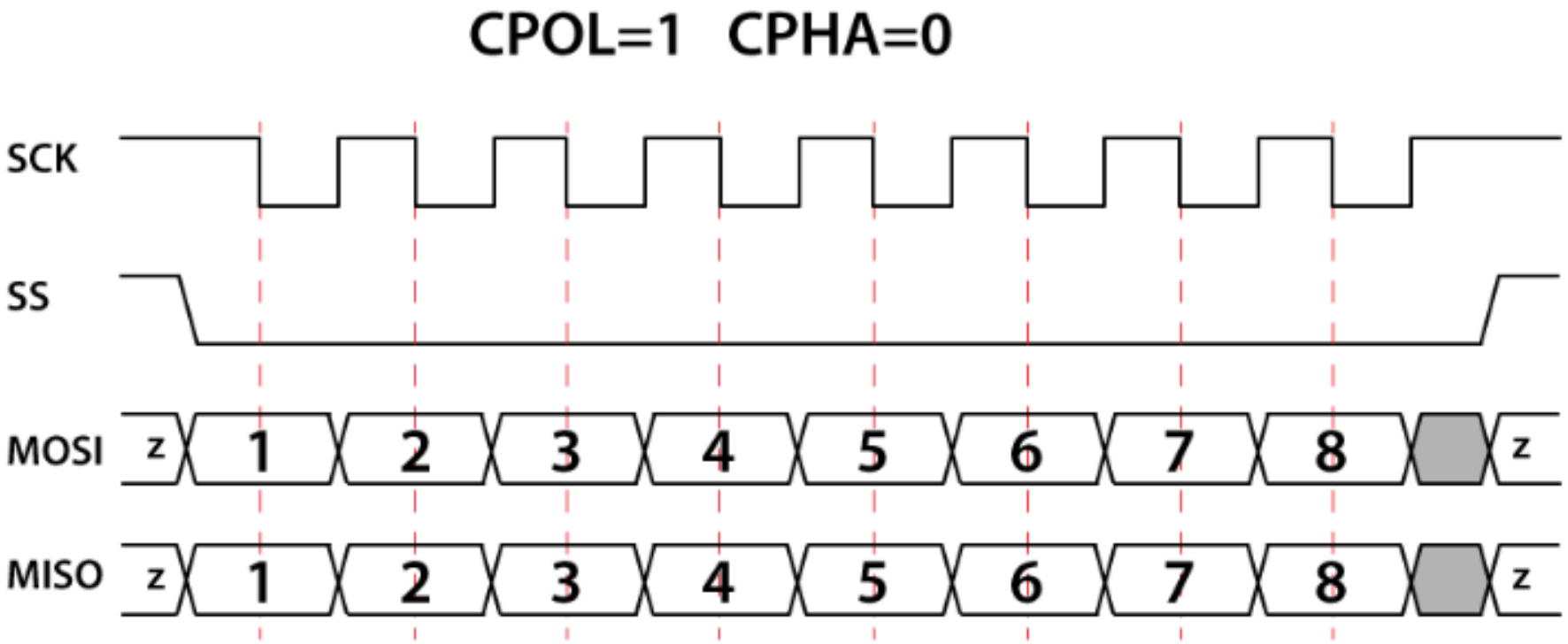
For **CPOL**=0, the base value of the clock is zero. For **CPHA**=0, data are captured on the clock's rising edge and data are propagated on a falling edge.

1.3.2. CPOL = 0, CPHA = 1



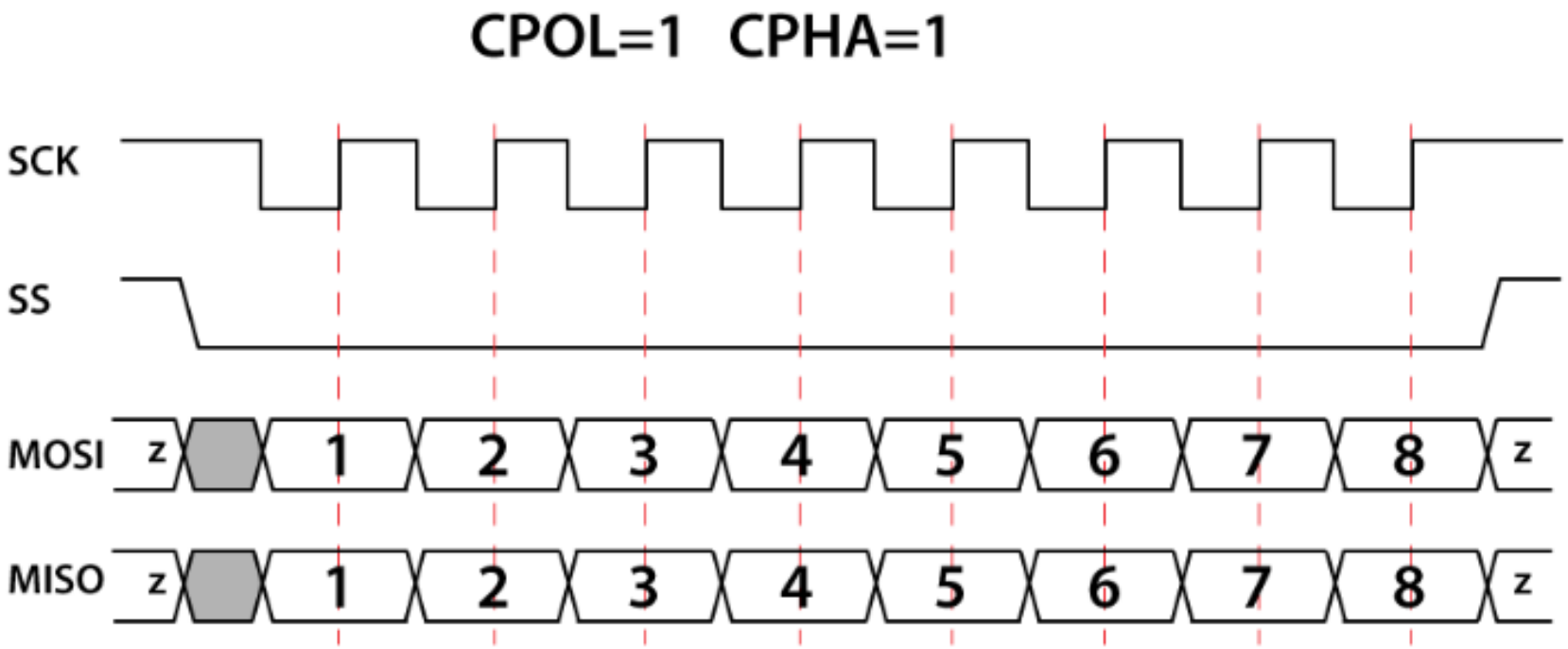
For **CPOL**=0, the base value of the clock is zero. For **CPHA**=1, data are captured on the clock's falling edge and data are propagated on a rising edge

1.3.2. CPOL = 1, CPHA = 0



For **CPOL**=1, the base value of the clock is one. For **CPHA**=0, data are captured on the clock's rising edge and data are propagated on a falling edge.

1.3.2. CPOL = 1, CPHA = 1



For **CPOL**=1, the base value of the clock is one. For **CPHA**=1, data are captured on the clock's falling edge and data are propagated on a rising edge

2. LPSPI Overview

LPSPI - **Low Power Serial Peripheral Interface**

The LPSPI supports:

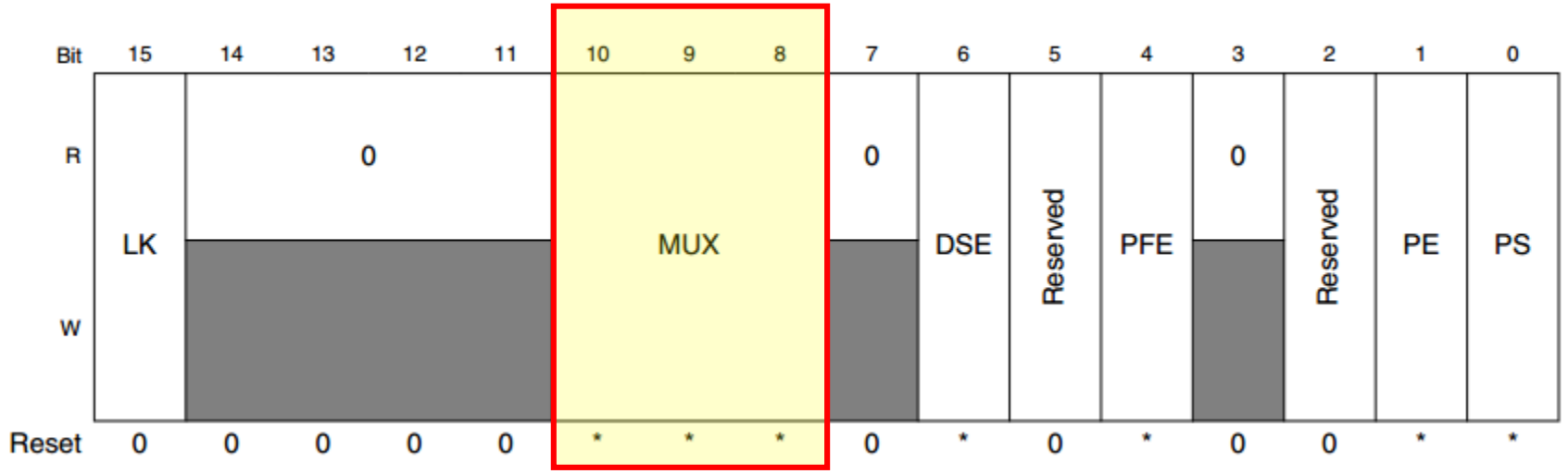
- Word size = 32 bits
- Configurable clock polarity and clock phase
- Master operation supporting up to 4 peripheral chip selects
- Slave operation
- Command/transmit FIFO of 4 words
- Receive FIFO of 4 words
- Flexible timing parameters in master mode
- Receive data match logic supporting wakeup on data match

2.1. LPSPI signals

Table 51-3. Signals

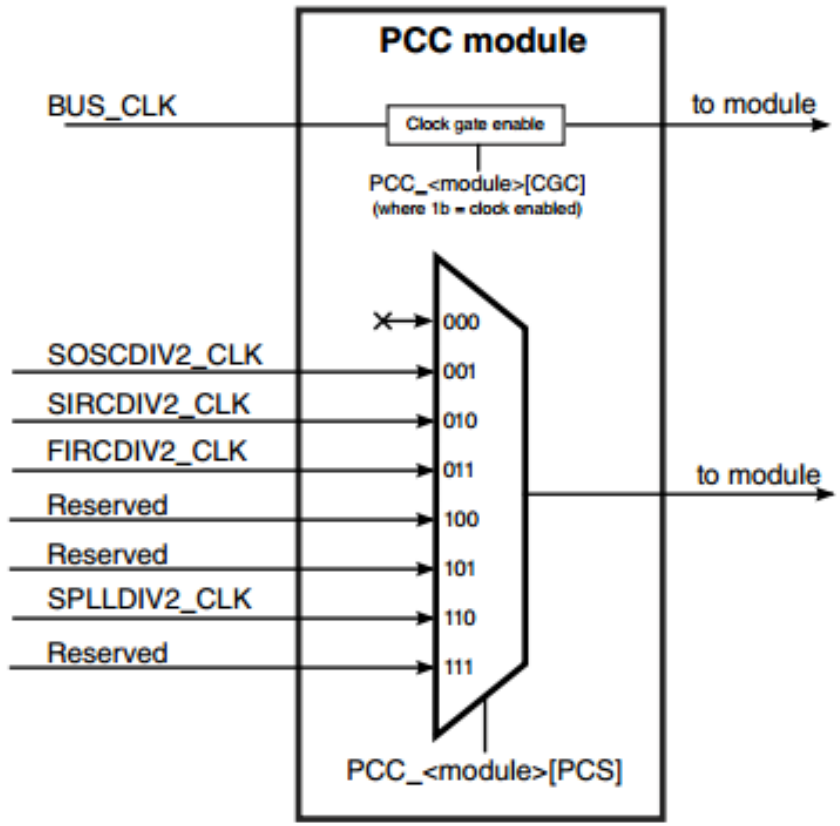
Signal	Name	Description	I/O
SCK	Serial clock	• Input in slave mode • Output in master mode	I/O
PCS[0]	Peripheral Chip Select	• Input in slave mode • Output in master mode	I/O
PCS[1] / HREQ	Peripheral Chip Select or Host Request	Host Request pin is selected when HREN=1 and HRSEL=0 • Input in either slave mode or when used as master Host Request • Output in either master mode	I/O
PCS[2] / DATA[2]	Peripheral Chip Select or data pin 2 during parallel data transfers	• Input in slave mode • Output in master mode • Input in half-duplex parallel data receive transfers • Output in half-duplex parallel data transmit transfers	I/O
PCS[3] / DATA[3]	Peripheral Chip Select or data pin 3 during parallel data transfers	• Input in slave mode • Output in master mode • Input in half-duplex parallel data receive transfers • Output in half-duplex parallel data transmit transfers	I/O
SOUT / DATA[0]	Serial Data Output	Can be configured as serial data input signal • Used as data pin 0 in half-duplex parallel data transfers	I/O
SIN / DATA[1]	Serial Data Input	Can be configured as serial data output signal • Used as data pin 1 in half-duplex parallel data transfers	I/O

2.2. Setting SCK/PCS/SOUT/SIN pin

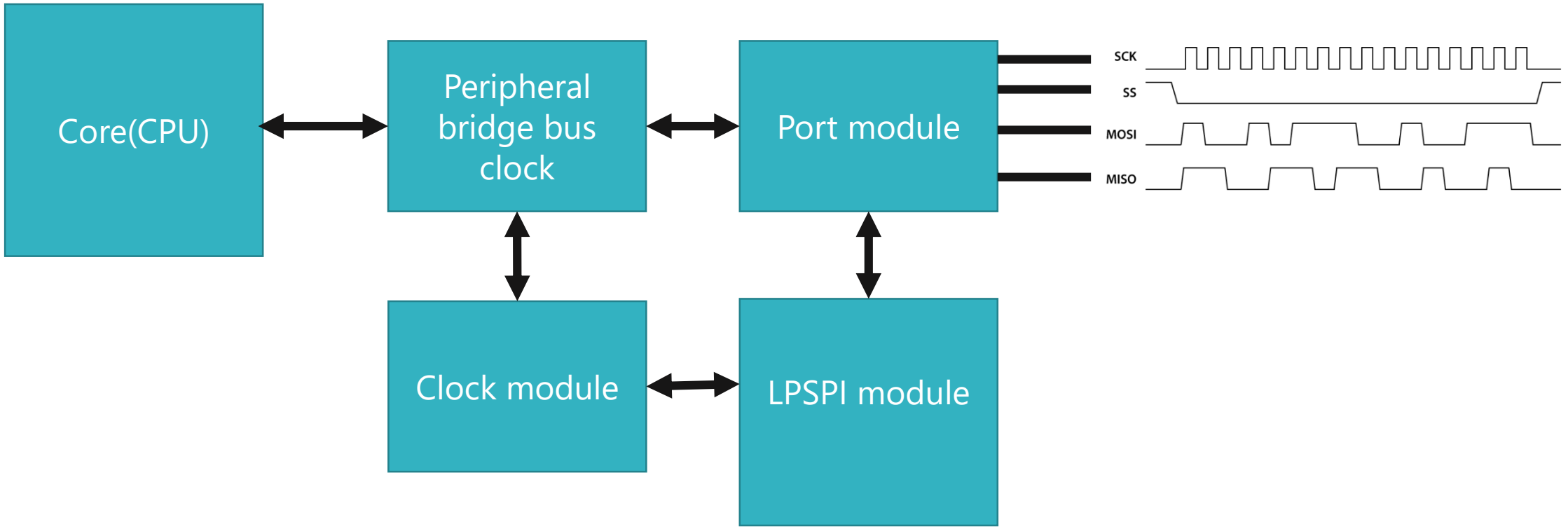


10–8 MUX	Pin Mux Control Not all pins support all pin muxing slots. Unimplemented pin muxing slots are reserved and may result in configuring the pin for a different pin muxing slot. The corresponding pin is configured in the following pin muxing slot as follows: 000 Pin disabled (Alternative 0) (analog). 001 Alternative 1 (GPIO). 010 Alternative 2 (chip-specific). 011 Alternative 3 (chip-specific). 100 Alternative 4 (chip-specific). 101 Alternative 5 (chip-specific). 110 Alternative 6 (chip-specific). 111 Alternative 7 (chip-specific).

2.3. Source clock LPSPI

Modules	PCC multiplexer
LPSPI	

2.3. Source clock LPSPI



2.4. Block diagram

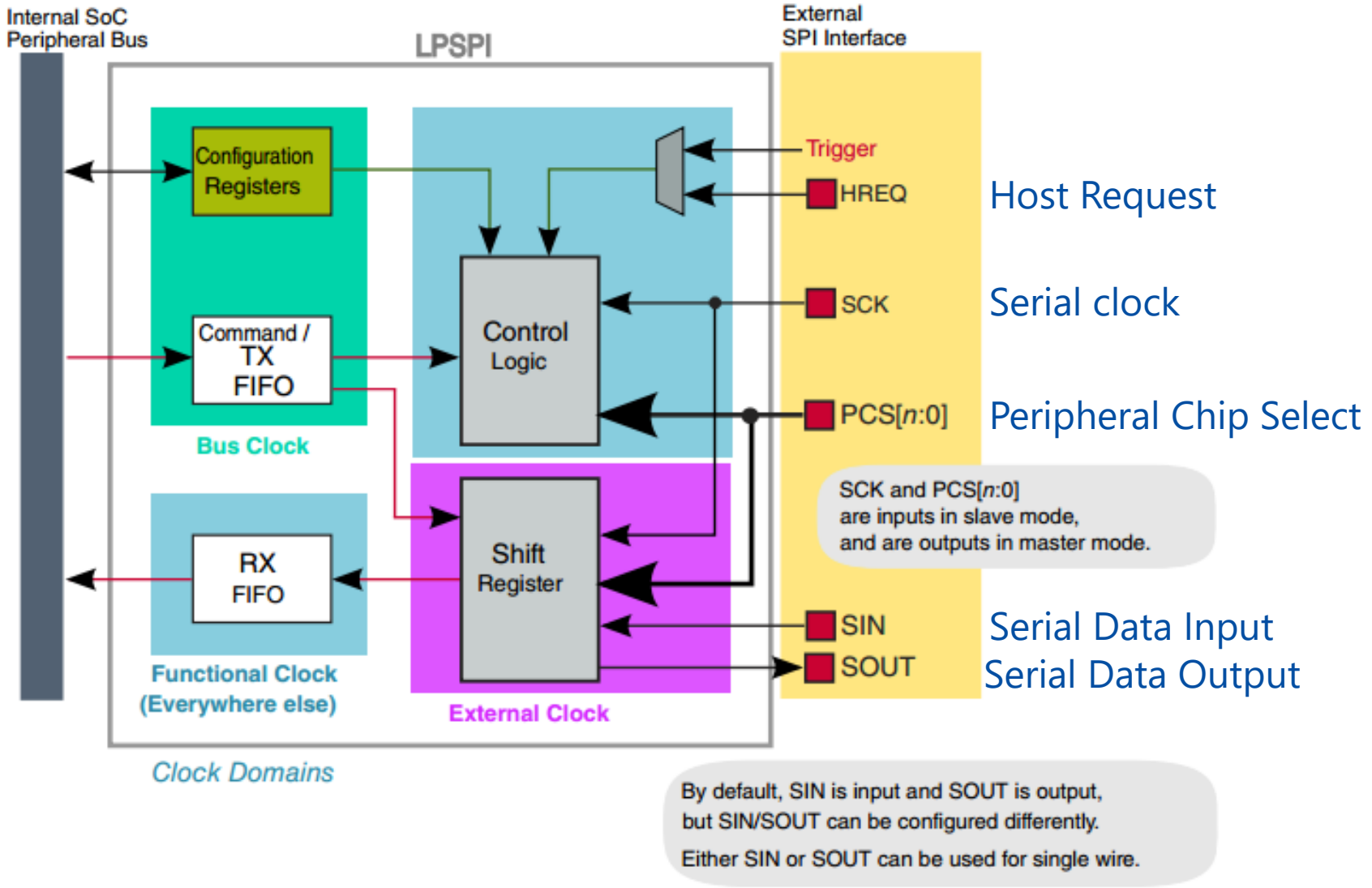
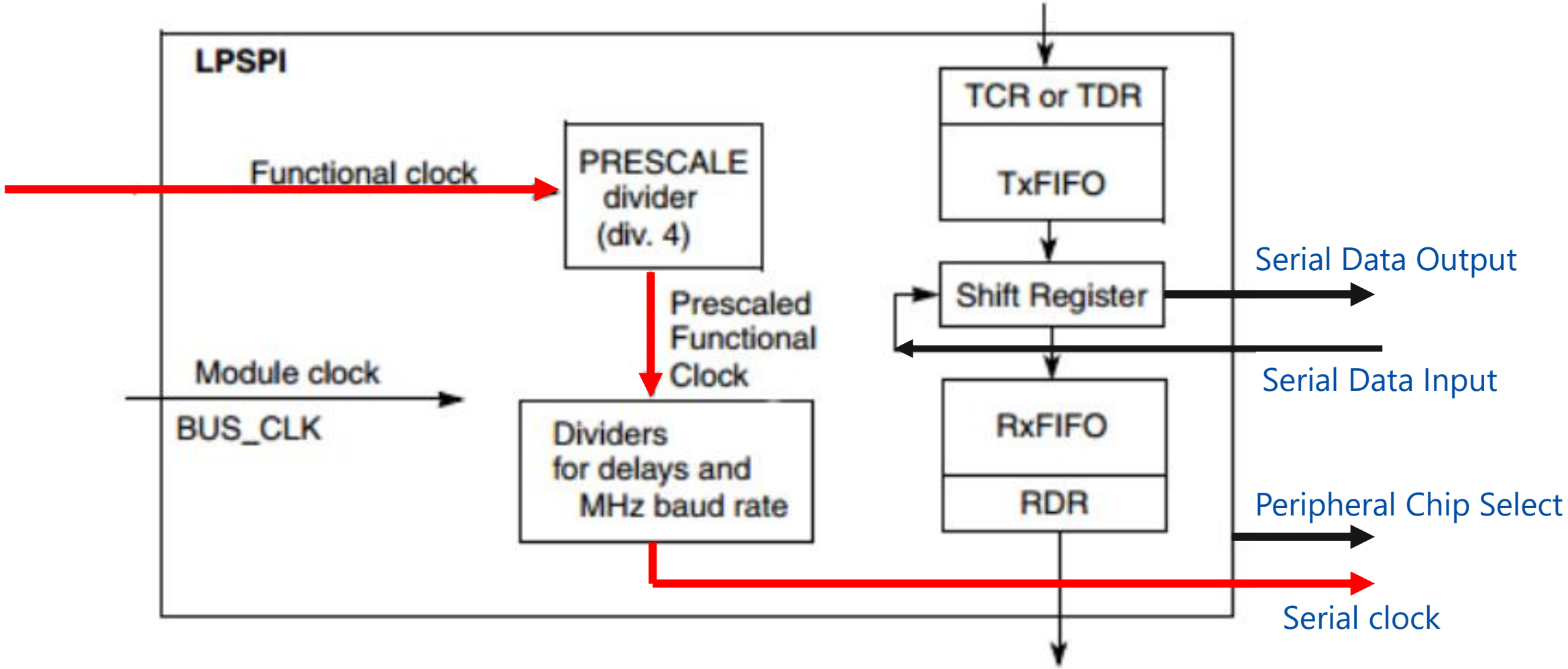


Figure 51-1. Block Diagram

2.4. Block diagram



2.5. LPSPI memory map

51.3.1.1 LPSPI memory map

LPSPI0 base address: 4002_C000h

LPSPI1 base address: 4002_D000h

LPSPI2 base address: 4002_E000h

Offset	Register	Width (In bits)	Access	Reset value
0h	Version ID Register (VERID)	32	RO	0100_0004h
4h	Parameter Register (PARAM)	32	RO	0000_0202h
10h	Control Register (CR)	32	RW	0000_0000h
14h	Status Register (SR)	32	W1C	0000_0001h
18h	Interrupt Enable Register (IER)	32	RW	0000_0000h
1Ch	DMA Enable Register (DER)	32	RW	0000_0000h
20h	Configuration Register 0 (CFGR0)	32	RW	0000_0000h
24h	Configuration Register 1 (CFGR1)	32	RW	0000_0000h
30h	Data Match Register 0 (DMR0)	32	RW	0000_0000h
34h	Data Match Register 1 (DMR1)	32	RW	0000_0000h
40h	Clock Configuration Register (CCR)	32	RW	0000_0000h
58h	FIFO Control Register (FCR)	32	RW	0000_0000h
5Ch	FIFO Status Register (FSR)	32	RO	0000_0000h
60h	Transmit Command Register (TCR)	32	RW	0000_001Fh
64h	Transmit Data Register (TDR)	32	WO	0000_0000h
70h	Receive Status Register (RSR)	32	RO	0000_0002h
74h	Receive Data Register (RDR)	32	RO	0000_0000h

2.5.1. Assignment 1

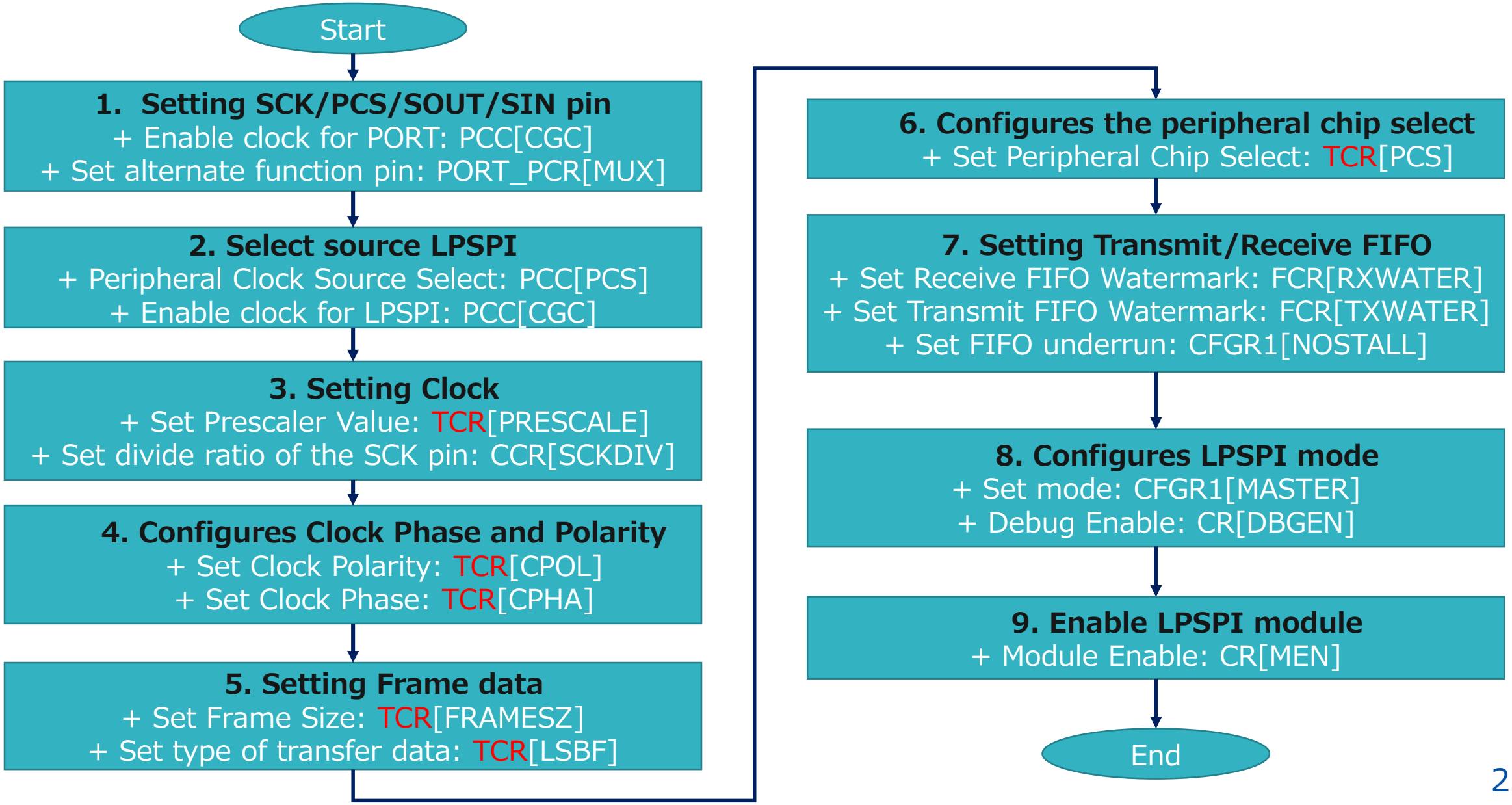
Write a define struct to manage LPSPI memory map

Ex:

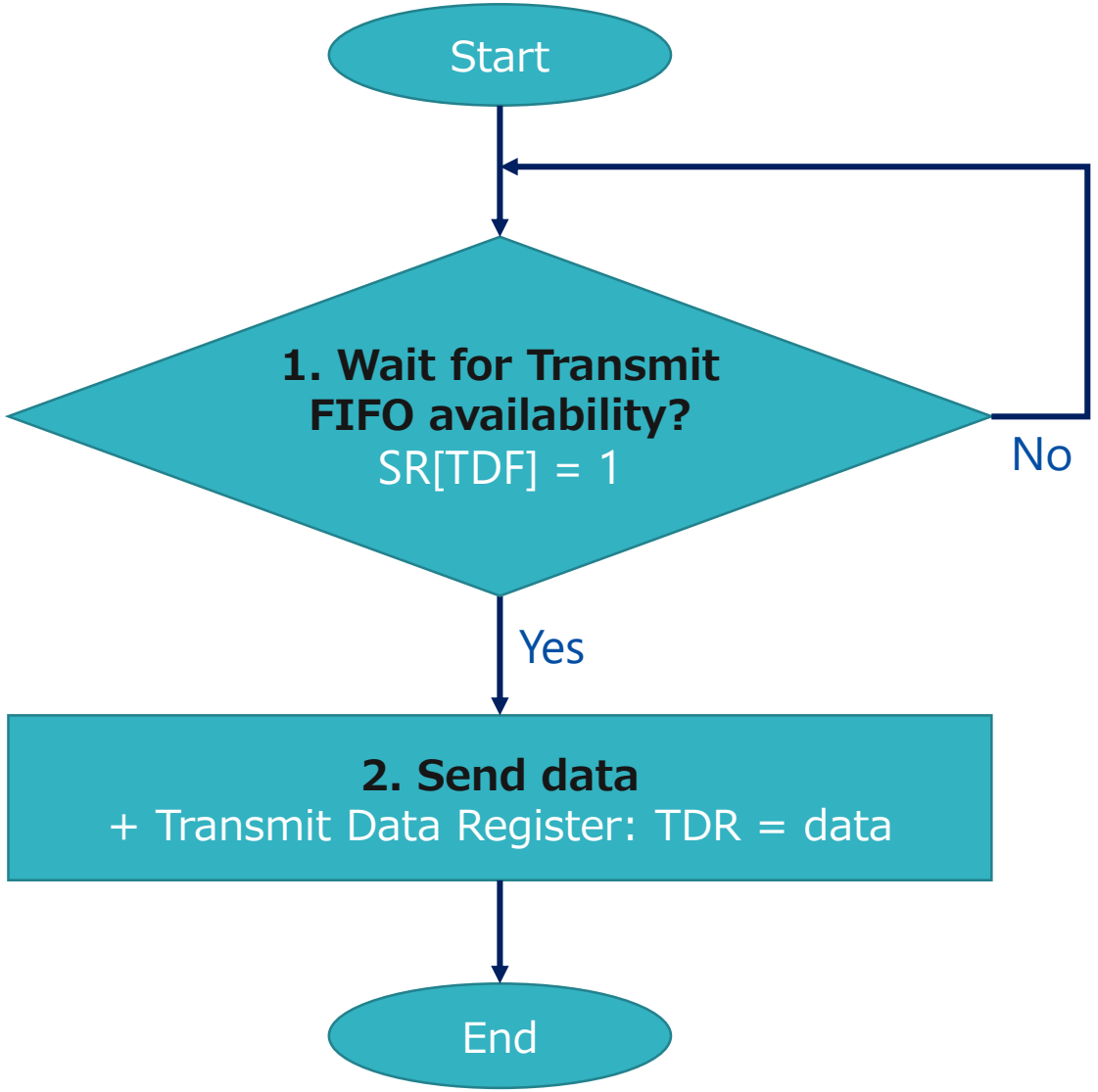
```
typedef struct {
        register
        ……
} LPSPI_Type;

#define LPSPI0_base_address (0x4002C000u)
#define LPSPI ((LPSPI_Type *) LPSPI0_base_address)
```

2.6. Configuration for LPSPI



2.7. Configuration for LPSPI transmit data



3.1. Assignment 2

Write a program to **Transmit** single char via LPSPI

- CLK = 1MHz
- CPOL, CPHA = 0
- 8 bit format
- Send char: ‘h’

3.2. Assignment 3

Write a program to **Transmit** a string via LPSPI

- CLK = 1MHz
- CPOL, CPHA = 0
- 8 bit format
- Send char: “Hello world!”


Thank you